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Auto-support: traction force constrained support structures optimization for bottom-up vat photopolymerization

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ABSTRACT

Bottom-up vat photopolymerisation offers high precision and speed, but the process is influenced by separation forces, which vary with the cross-sectional area of printed layers—often overlooked in existing support structure generation methods. Insufficient supports lead to printing failures, while excessive supports require manual intervention and degrade surface quality. Optimising support structures to account for separation forces is challenging due to the computational expense of repeated simulations. This paper introduces a novel methodology to optimise support structures, minimising support usage while satisfying force and overhang constraints. The approach combines representative layer simulation and geometric optimisation with simulation verification, reducing simulation iterations from hundreds of thousands to a few dozen. Layers are clustered by cross-sectional shape coherence, with representative layers showing the maximum support traction forces identified for full simulations. These simulations guide and validate geometric optimisation to meet both overhang and force constraints. Experimental results show that this approach significantly reduces support contact points compared to commercial software while maintaining structural stability throughout the printing process.

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Support structures; vat photopolymerization; separation force; support optimization

1. Introduction

Vat photopolymerization is widely used in applications such as metamaterials [1, 2], jewellery [3], footwear [4], dentistry [5], and biomedical engineering [6, 7], due to its high precision and ability to create complex geometries with superior surface quality.

The bottom-up vat photopolymerization technique reduces material consumption by using less resin for taller models [8]. As the build platform ascends, a vacuum forms, refilling the vat with resin and eliminating the need for a wiper to ensure layer accuracy. This process enables high-resolution Z-axis printing and improves model quality by reducing surface tension. However, it introduces challenges with separation forces during ascent, which can damage support structures as the cured layer detaches from the resin vat.

Support structures are crucial in anchoring printed parts to the platform and counteracting separation forces that could otherwise cause structural failures. The support generation process must balance several conflicting requirements. Support structures must be strong enough to withstand the load during separation [9] yet minimal enough to reduce post-processing effort and prevent surface damage [10–13] Figure 1.

Specifically, inadequate support structures can result in two primary types of printing failures: (1) Failure to satisfy overhang constraints, where insufficient support for the lowest points or overhanging areas leads to sagging or collapse during printing, and (2) Failure to satisfy force constraints, where the traction force fracture threshold of the supports is less than the forces required to counteract the shape's gravity and separation forces during separation along the Z-axis, causing support deformation or fracture and ultimately print failure. Therefore, generating support structures that can sustain these loads is critical for successful bottom-up vat photopolymerization printing.

Current methods often focus on overhang constraints, typically adding uniformly distributed support points to address these issues [14–16]. Wang et al. [17] and Chen et al. [18] modelled the support addition

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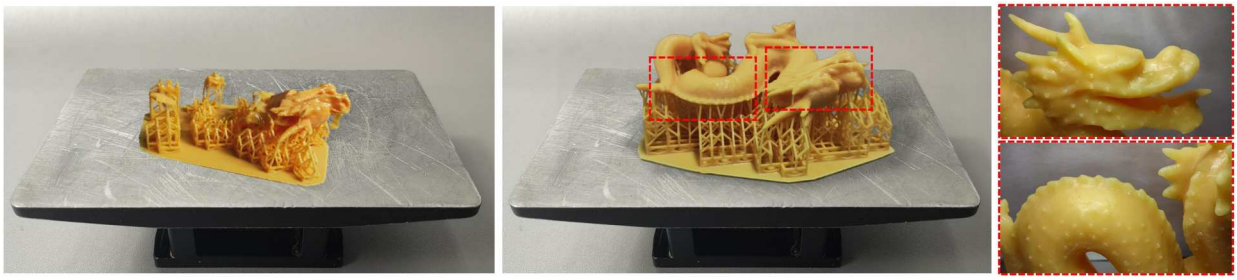


Figure 1. Effects of support number on print quality.

problem as a circle-covering problem, optimising support point positions to reduce both the number of supports and their total height. However, these methods often neglect load considerations. Some researchers, such as Jang et al. [19] considered the gravity of the model, adding support points in unsupported areas to reduce the risk of breakage. Topology optimisation methods [20–22] have also been employed to design supports that account for gravity or residual stress. However, these approaches do not consider separation forces, which are critical in bottom-up vat photopolymerization.

Zhao et al. [23] optimised support radii based on structural loads. However, their method only considered single-layer separation forces using simplified distributions rather than simulations, hindering its practical implementation. Moreover, their approach generated large-radius supports that caused removal difficulties and more surface damage [24]. Materialise e-Stage 7.3 improves this by using fixed-radius supports with variable density, enabling easier removal with less surface damage. However, being closed-source, its principles remain unknown. Our results show that its supports may fail for small overhang regions, and even successful prints leave room for support quantity optimisation.

Optimizing support structures for bottom-up vat photopolymerization requires careful consideration of both load and overhang constraints to ensure structural integrity while minimising the number of support points. This optimisation process presents two significant challenges. First, extensive simulations are necessary, which can be time-consuming. Second, the optimisation must balance force and overhang constraints to minimise the number of support points.

During printing, the distribution of traction forces on supports varies with cross-sectional size and can differ even among identical cross-sections with varying geometries, as demonstrated in Figure 2. This uncertainty in identifying which layer will impose the maximum traction necessitates simulating the separation process for each layer. For models with hundreds to thousands of layers, an equal number of simulations are required to assess the separation forces. When optimising support structures using topological optimisation methods, dozens to hundreds of iterations may be needed [21, 22], resulting in potentially hundreds of thousands of simulations. These high-frequency simulations require millisecond-level speeds, but for models with tens of thousands of elements, separation process simulations using Abaqus on a 24-core processor can take tens of seconds, which is relatively slow. Although machine learning techniques [25] and GPU-accelerated Partial Differential Equations (PDEs) solutions [26, 27] have been explored to enhance

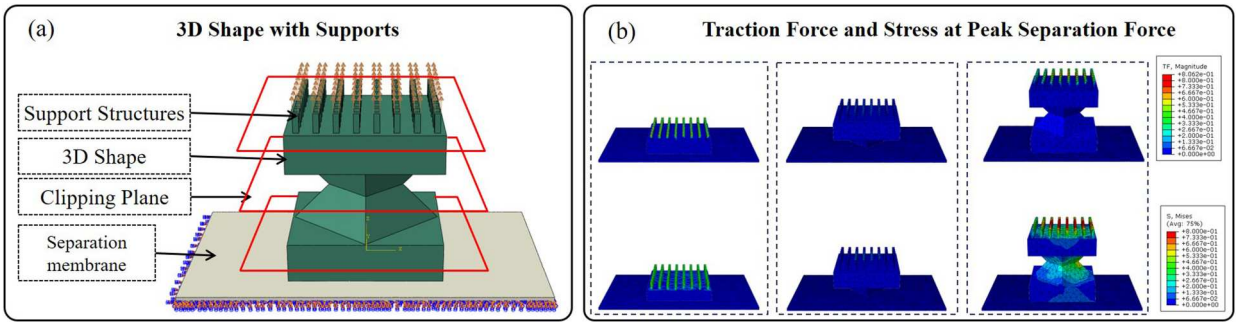


Figure 2. The schematic demonstrates the separation process for a shape with supports across different layers.

speeds, achieving sub-second simulations for complex models with tens of thousands of elements remains a significant challenge.

To minimise the number of support points, it is crucial to balance the traction forces across the supports, thereby reducing the total quantity needed. However, stress field-based methods, while ensuring structural strength, may miss necessary supports in overhanging areas, as shown in Figure 3. Balancing overhang constraints with the goal of minimising support points can be framed as a circle covering problem, which is NP-hard [17]. The optimisation process becomes more complex when balancing variably distributed support points, which align with the traction force field, against uniformly distributed points needed to meet overhang requirements.

The main contributions of this work are as follows:

- We tackle support structure optimisation in bottom-up vat photopolymerization by incorporating traction forces and propose a computationally practical optimisation framework, solving the distribution of support points as a capacity-constrained circle covering problem.
- We introduce a strategy that clusters layers based on cross-sectional shape coherence and applies simulations only to representative layers, which cause the maximum traction forces on the supports, significantly reducing the total number of simulations required for computing traction forces.
- We define force-augmented overhang regions that incorporate traction forces, enabling adaptive support distribution based on force conditions, effectively addressing issues of superfluous or insufficient support structures driven solely by geometric overhang.

2. Material and methods

2.1. Method overview

This section presents an adaptive support framework for bottom-up vat photopolymerization, consisting of input, output, and two core algorithm modules, as illustrated in Figure 4.

In the input phase, users provide the 3D shape and printing parameters. The 3D shape is then sliced according to the specified layer thickness. The representative layer extraction module clusters these sliced layers based on image similarity and index continuity, selecting the layer experiencing the maximum load within each cluster as the representative layer.

In the support structures optimisation module, the process begins by generating initial supports that address only the overhang constraints. Separation simulations are conducted on the representative layers to determine the support traction forces. This data guides the geometric optimisation of support positions and quantities until the force constraints are satisfied.

The final output is a set of optimised support structures that ensure the printed object is both structurally sound and efficiently supported.

2.2. Representative layers extraction

In this section, we present a novel clustering-based approach for support structure analysis in bottom-up vat photopolymerization, driven by the challenge of accurately and efficiently obtaining stress distributions for arbitrary 3D geometries during the separation process. Rather than attempting to predict stress distributions directly, which remains a difficult problem, we propose to group layers that induce similar loads on

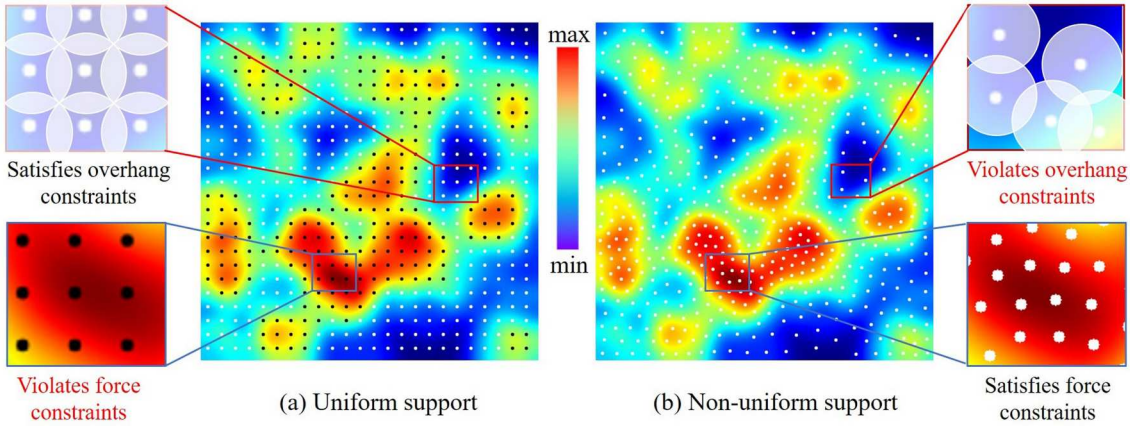


Figure 3. Support distributions governed by overhang constraints and traction force. A random stress field is visualized using a color gradient from blue (low stress) to red (high stress). (a) In the uniform support distribution, insufficient supports in high-stress regions violate force constraints, risking structural failure. (b) In the non-uniform distribution, supports in low-stress regions fail to satisfy overhang constraints, as circles with maximum overhang radius centered at supports cannot fully cover the overhang area.

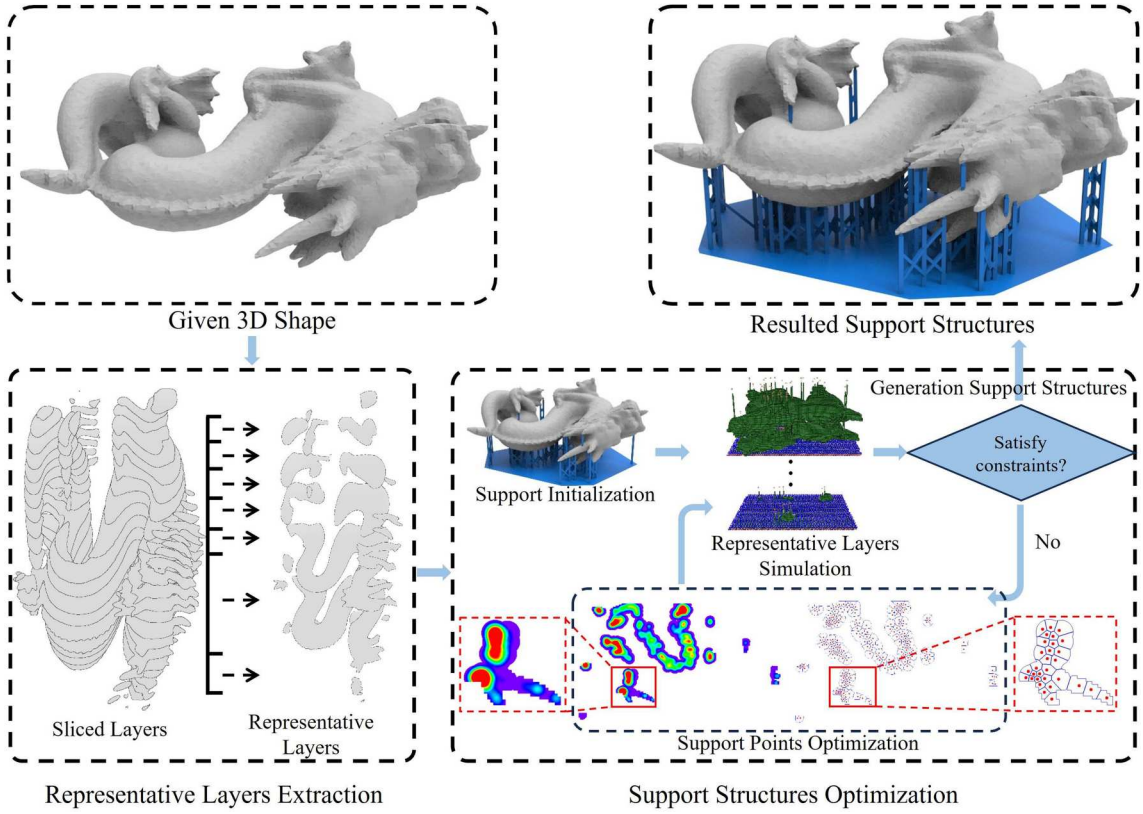


Figure 4. Adaptive support algorithm pipeline. Given a 3D shape and printing parameters, the Representative Layers Extraction module slices the model, clusters layers by image similarity, and selects the maximum load layer from each cluster for simulation. In the Support Structures Optimization phase, initial supports are generated to satisfy overhang constraints, and then support points are optimized iteratively until all constraints are met. The final support structures are produced based on the optimization results.

support structures into clusters and identify *representative layers* that experience maximum traction forces within each cluster. Through systematic analysis and a machine learning approach, we can efficiently identify these representative layers. By focussing simulations only on these critical layers, we significantly reduce computational costs while maintaining analysis accuracy and reliability.

Two primary factors influence the maximum traction force on supports: (1) the magnitude of the loads and (2) the three-dimensional geometry of the shape. The loads, which include separation forces and gravity, directly affect the traction force on the supports. In our experiments, separation forces can reach several hundred newtons, whereas gravity typically does not exceed 10 newtons. Consequently, the cross-section that generates the separation force is crucial. Additionally, the 3D geometry of the model affects the traction forces on the supports. As shown in Figure 2, although two shapes may have identical cross-sections, differences in the model's geometry lead to variations in stress and traction forces, even under the same separation force.

To address this, Section 2.2.1 details the clustering of adjacent layers with continuous indexing and similar cross-sections, ensuring consistency in both cross-section and 3D shape within each cluster. Section 2.2.2 describes our approach for efficiently identifying the layer within each cluster that experiences the maximum load. The maximum load layers from all clusters are then used as representative layers, optimising the simulation process for both efficiency and accuracy. For details on the representative layer extraction algorithm, refer to Algorithm 1.

Algorithm 1 Representative Layers Extraction

Input: Cross-sectional images *Img* with printing order
Output: Representative layer indices C_l
// Cross-Section Clustering
Downsample, binarize, and normalize *Img*;
Calculate distances between adjacent clusters and record the minimum distance d_{min};
while $d_{min} < \epsilon_d$ **do**
  Merge the two nearest clusters;
  Update distances between the merged cluster and adjacent clusters;
  Recalculate and update d_{min};
end while
// Identify Maximum Load Layer per Cluster
For each cluster, identify the layer with the maximum load and add it to C_l;
Return: C_l

2.2.1. Clustering of adjacent and similar layers

This section outlines a clustering method that iteratively merges adjacent and similar layers until a predefined threshold is reached, resulting in multiple categories that ensure contiguous indices with similar cross-sections.

Before clustering, the sliced images undergo preprocessing: they are first downsampled to reduce resolution and dimensionality, then binarized to simplify the data into black and white, and finally normalised so that all pixel values are either 0 or 1.

Given that models may contain both large and small cross-sections, it is important to carefully define the distance function and threshold. Using Euclidean distance alone as a similarity measure can make it difficult to determine an appropriate distance threshold for clustering. Therefore, we define the distance function as a relative distance:

$$d_c = \frac{d(i, j)}{\max(\|i\|, \|j\|)}, \tag{1}$$

where $d(i, j)$ represents the Euclidean distance between images i and j, and $\|i\|$ is the sum of the pixel values in image i.

To maintain the continuity of slice image indices within each cluster, we employ the following clustering steps: If the image indices between two classes are not consecutive, their distance is set to be greater than a preset threshold, preventing the merging of non-adjacent images. For consecutive indices, the distance is defined as the maximum distance between slices within those clusters, as described by Equation (1). This method effectively groups similar and adjacent slices into clusters with continuous indices.

For determining the optimal distance threshold ϵ_d, we adopted the method of Rahmah et al. [28], which examines the distance distribution between slice images. This approach derives the Eps value by analysing the sorted k-distance graph, where a significant change in slope indicates the optimal threshold for clustering.

2.2.2. Identifying the maximum load layer per cluster

This section describes our method for quickly identifying the layer within each cluster that experiences the maximum load. Determining separation forces is a complex task, so our research focuses on calculating the Z-direction separation forces to set the slicing positions of the model, rather than analysing detailed stress distributions. We simplify the problem by disregarding the influence of the printed layer shapes on stress distribution. The primary factor affecting the separation force is the shape and region of the vacuum [29, 30], rather than the printed layers themselves. We use sliced images to predict the Z-direction separation force needed to break the vacuum for each layer, offering a more efficient approach compared to using 3D models. This method reduces data complexity, speeds up dataset generation, and increases efficiency in training and inference due to lower computational demands.

To verify that the printed shape affects only the stress distribution and not the separation force, we conducted simulations shown in Figure 5. In those simulations, the models on the top and bottom have identical printing cross-sections but different stress distributions. These differences arise from variations in the overall 3D shape, leading to distinct and unpredictable stress distributions even for identical cross-sections. However, despite these variations, the Z-direction separation

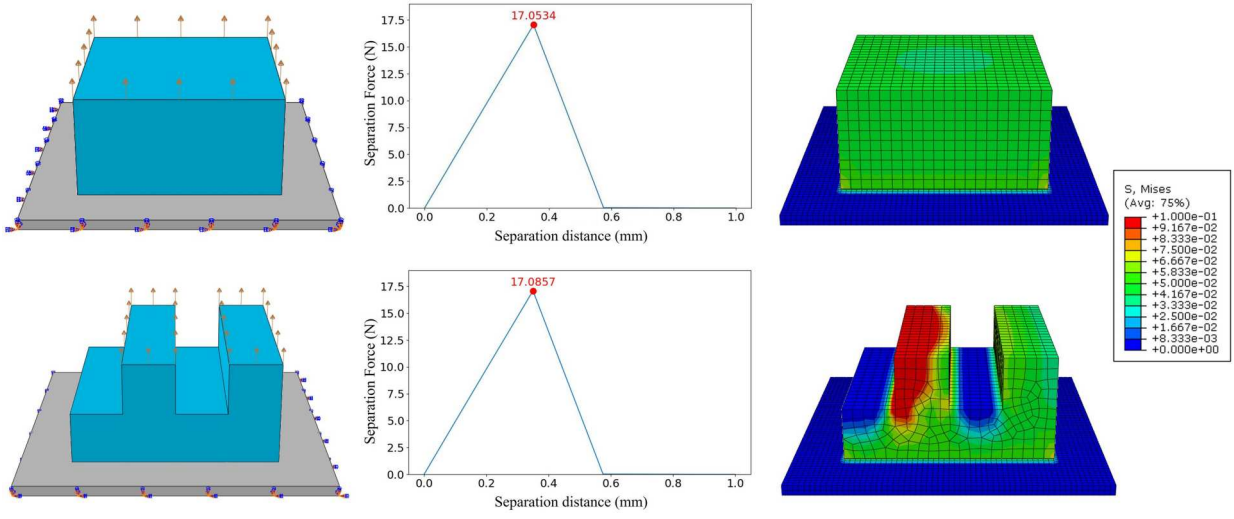


Figure 5. Comparison of zero-gravity simulation results for two models with identical cross-sections but different 3D shapes.

forces remain almost the same, with a difference of only 0.17%, which falls within the margin of simulation error.

The image resolution should match the printable range of the printer, with each pixel representing a millimetre. In this study, we use 192 × 192 pixel images as input, representing the cross-sectional slice of the current printing layer. The image is unfolded from left to right, top to bottom, into a one-dimensional vector with 192 × 192 elements, which is then fed into a multilayer perceptron (MLP) network. The network outputs a scalar value representing the Z-direction traction force exerted on that slice, as illustrated in Figure 6. This approach converts the separation force prediction into a regression problem, with the MLP network mapping the relationship between the slice image and the separation force.

The MLP is a fully connected neural network comprising input, hidden, and output layers, where each layer is fully connected to the previous and next layers. The output of each neurone is a weighted sum followed by a nonlinear activation function, allowing for nonlinear fitting. As shown in Figure 6, the activation value $a^{(i)}$ of the ith layer is passed as input to the $(i+1)$th layer, with the output given by:

$$a^{(i+1)} = \phi(w^{(i)}a^{(i)}), \tag{2}$$

where i denotes the current layer number, $w^{(i)}$ represents the weights of the ith layer, and ϕ is the ReLU activation function:

$$ReLU{:}f(x) = \max(0, x). \tag{3}$$

In addition to the predicted separation force, we also account for the gravitational force of the printed parts when determining the force exerted on each layer slice. To calculate the gravity of the printed model, we first multiply the area of each layer slice by the slice thickness to obtain the volume of each layer. Summing the volumes of all printed layers gives the total volume of the printed model. Finally, multiplying the total volume by the material density yields the gravitational force of the printed model.

2.3. Support structures optimisation

Optimizing the placement of support points improves the uniformity of traction force distribution and maximises the load-bearing of each support point, thereby reducing the overall number of supports needed. Section 2.3.1 details the simulation methods used to analyse forces on supports after segmenting representative layers. While supports are typically placed in overhang regions, previous definitions [18] of these regions did not account for separation forces. Therefore, we redefine overhang regions in Section 2.3.2. Section 2.3.3 introduces a geometric optimisation process to adjust the placement and quantity of support points, using simulations primarily for verification and fine-tuning of the stress field to reduce the number of simulations required.

2.3.1. Representative layers simulation

In this section, we simulate the separation process on representative layers using dynamic simulation with the boundary conditions shown in Figure 7. The Z-direction movement speed of the support structure is set to 2 mm/s, while the bottom of the separation film is fixed. The 3D shape and the top of the separation film are set in contact, employing the CZM model [31] for simulating their separation. The separation force is calculated using:

$$t = \begin{Bmatrix} t_n \\ t_s \\ t_t \end{Bmatrix} = \begin{bmatrix} K_{nn} & K_{ns} & K_{nt} \\ K_{ns} & K_{ss} & K_{st} \\ K_{nt} & K_{st} & K_{tt} \end{bmatrix} \begin{Bmatrix} \delta_n \\ \delta_s \\ \delta_t \end{Bmatrix} = K\delta, \tag{4}$$

where t represents the traction vector, K is the stiffness matrix, and δ is the separation distance matrix.

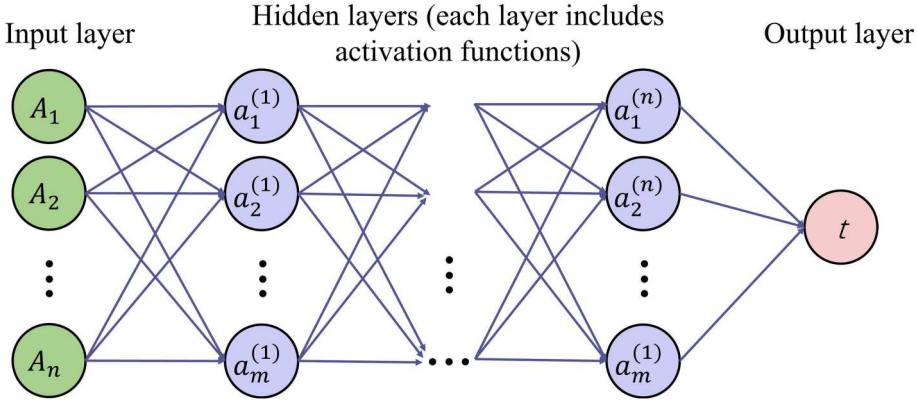


Figure 6. Diagram of the Multi-Layer Perceptron network architecture.

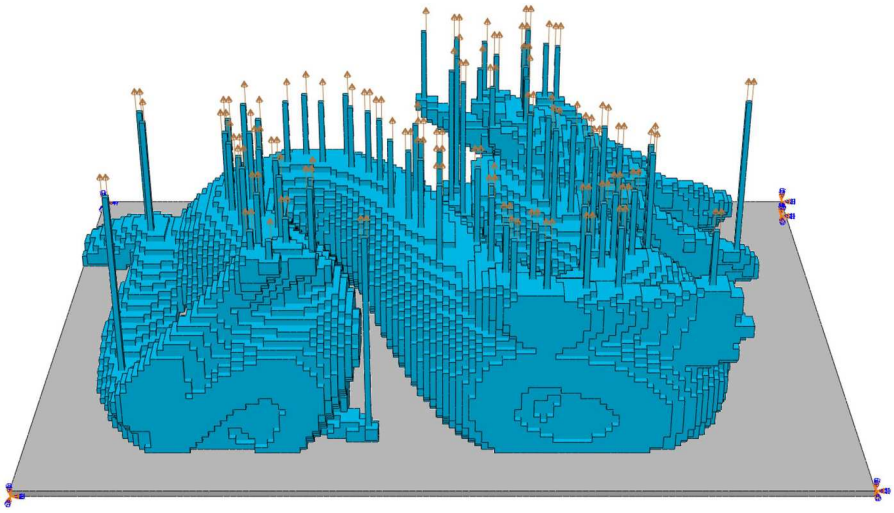


Figure 7. Simulation boundary conditions: A uniform velocity is applied to the top endpoint of the support in the positive Z-axis direction, along with the bottom of the separation film.

To ensure simulation stability and efficiency, we use voxel grids for mesh division. Voxel grids allow rapid mesh generation and simplify updates by only adjusting affected voxels when support positions or numbers change, eliminating the need for full re-meshing. The voxel grid size is set to match the cross-sectional dimensions of the support structure.

In our finite element model Figure 7, we omitted auxiliary truss structures for shear force resistance, as the mechanics of bottom-up photopolymerization printing involves primarily Z-directional forces (separation and gravitational), where support failure is dominated by axial traction. Finite element analysis confirmed this simplification Figure 8, revealing nearly identical stress distributions in support rods (load variations less 2%) between truss and truss-free configurations, and truss elements demonstrate an order of magnitude lower stress than their connected supports. Truss structure inclusion would demand significantly finer voxel discretization, increasing computational

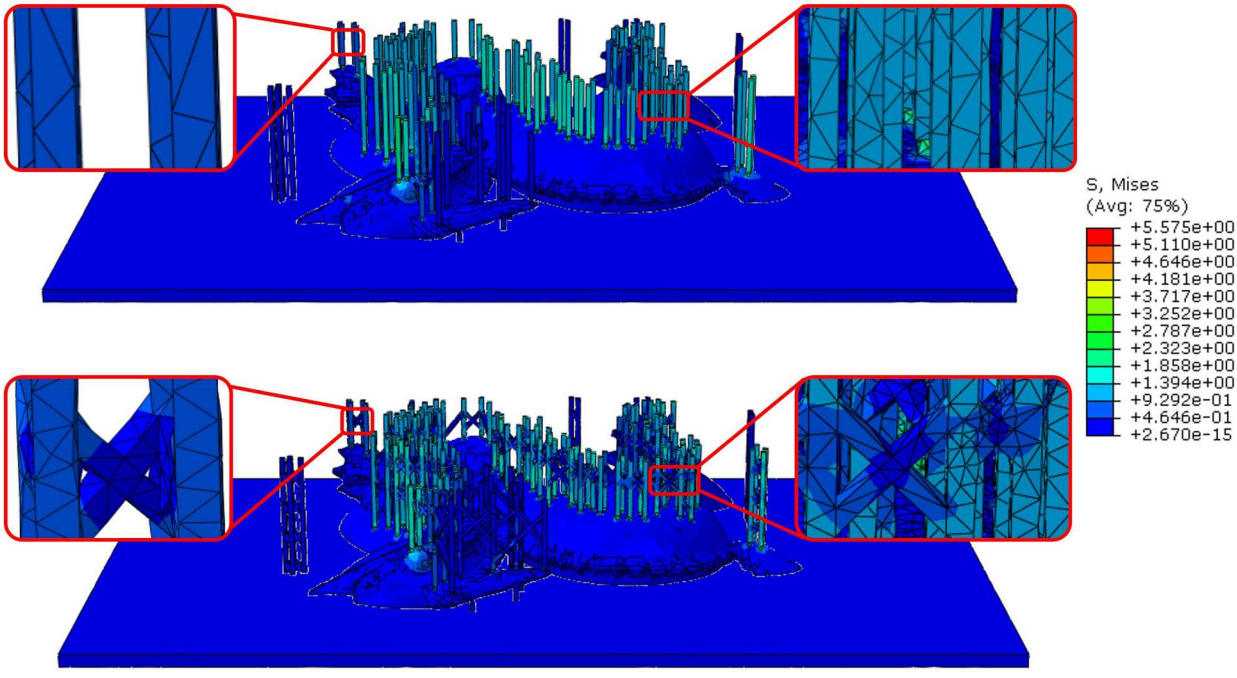


Figure 8. Stress analysis comparing support structures with and without truss reinforcement. The nearly identical stress distributions in support rods across both configurations validate the truss-free simulation approach, while the truss structure exhibits stress levels an order of magnitude lower than its connected supports.



Figure 9. Comparison of support structures for a conical model.

overhead with marginal improvements in prediction accuracy.

2.3.2. Overhang regions

Supports are typically added in overhang regions, but defining these regions effectively remains challenging. While conventional methods identify overhang regions based solely on geometric angles, they either provide insufficient support by being too strict or generate excessive supports by being too lenient. As demonstrated in Figure 9, in the case of an inverted conical model, this limitation leads to either inadequate support that can cause fractures (left, e-Stage with overhang angle$<$ 70°), or unnecessary supports in areas that may not require them (middle, overhang angle$\geq$ 70°).

To address these shortcomings, this paper redefines overhang regions in bottom-up vat photopolymerization, classifying them into 'geometric overhang regions' and 'force-augmented overhang regions'.

Geometric overhang regions refer to areas with significant overhangs that necessitate support structures to prevent printing failures. These regions are characterised by either a small angle relative to the build platform or an overhang length that exceeds the material's self-supporting capabilities, as defined in traditional methods.

Force-augmented overhang regions, on the other hand, are areas where additional supports are required beyond those in geometric overhang regions, particularly when the support structures in the latter are insufficient to meet force constraints. Unlike geometric overhang regions, force-augmented overhang regions are not bound by overhang angle thresholds, as their primary function is to reinforce the overall support framework, as illustrated in Figure 9 (right).

The extraction methods for these two types of overhang regions are as follows.

Geometric overhang regions, in a mesh representation, are identified where the angle between the mesh face and the XOY plane is less than a critical angle α_c, and the dot product of the mesh face's normal with the positive Z-axis is negative (Figure 10, left). In a slice contour representation, they are defined as regions where the overhang distance between the current and previous layers exceeds the material's maximum self-support distance $d_{self\text{-}support}$ (Figure 10, right).

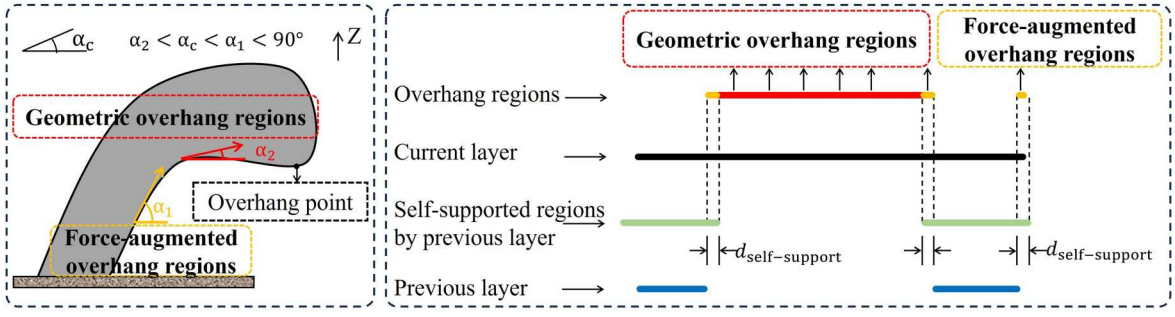


Figure 10. Determination of geometric overhang and force-augmented overhang regions in mesh (left) and slice (right) representations.

Force-augmented overhang regions, in a mesh representation, are defined where the angle between the mesh face and the XOY plane is greater than α_c but less than 90°, and the dot product of the mesh face's normal with the positive Z-axis is negative (Figure 10, left). In a slice contour representation, force-augmented overhang regions are those where the overhang distance between the current and previous layers is less than $d_{self\text{-}support}$ but greater than zero (Figure 10, right).

Since the constraints of geometric overhang regions are independent of the Z-axis, overhang regions are typically projected onto the XOY plane. The number and positions of supports are then optimised to satisfy coverage constraints [17, 32], or support points are generated on the XOY plane and projected onto the model [14]. However, if multiple overhang faces share the same (x, y) coordinates, direct downward projection may cause overlapping, as shown in Figure 11(left). A uniform distribution of supports in these overlapping regions allows for easier mapping back to the model's overhang areas. In contrast, a non-uniform distribution could lead to incorrect support placement. To address this issue, we propose dividing overlapping overhang regions into sub-regions and mapping them to distinct locations on the XOY plane, as depicted in Figure 11(right).

2.3.3. Adaptive support distribution algorithm

The support optimisation algorithm must satisfy two constraints: force constraints and overhang constraints. If the initial supports generated based on overhang constraints also meet the force constraints, the supports are instantiated directly. If not, the algorithm iteratively refines the number and positioning of support points through geometric optimisation and physical simulation until both constraints are concurrently satisfied.

Wang et al. [17] modelled the optimisation of support points under overhang constraints as a circle covering problem. In bottom-up vat photopolymerization printing, force constraints must also be considered alongside overhang constraints. Consequently, the problem is formalised as a 'Capacity-Constrained Circle Covering Problem', focussing on limiting the maximum force on supports under overhang constraints to minimise the number of supports. By mapping traction forces as a stress field and assuming that the nearest support point bears this stress, geometric optimisation is employed to balance the force on each support point within the traction force threshold, effectively substituting geometric optimisation for physical processes. This section will elaborate on the implementation of this process.

First, a uniform distribution of support points is initialised to satisfy overhang constraints using Equation (12), also referred to [33]. This method utilises the same formulas as the adaptive support, which will be detailed in the following text. Next, the traction force at each support point for every segmented model is obtained through simulation analysis. To ensure the integrity of the support structure throughout the printing process, the maximum traction force encountered

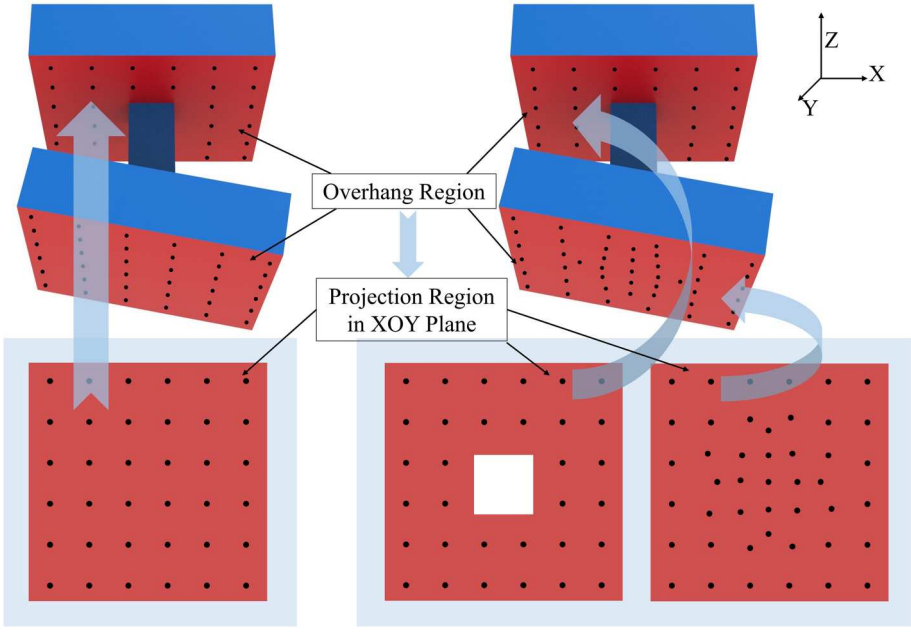


Figure 11. Schematic illustration of overhang region mapping methods. (a) Uniform support approach [14] maps multiple overhang regions to identical XOY plane locations. (b) Our non-uniform approach maps overhang regions to distinct locations for force-based support optimization before inverse mapping.

at each support point across all layers is selected as the traction force for that support point.

After determining the maximum traction force for each support point, further optimisation of the position and number of supports is performed. In regions where the traction force exceeds a support fracture threshold F_{break}, additional support points are added to prevent fracture; in regions with lower traction forces, the number of support points is reduced to conserve material and minimise surface damage to the model.

Instead of directly modifying the position and density of support points based on the traction forces they experience, the algorithm transforms the traction force on the supports into a stress field for the overhang regions. This transformation offers a more continuous and detailed perspective, guiding the distribution of supports more precisely. Specifically, the Gaussian distribution in Equation (5) is used to convert the traction force at the support points into a stress field, denoted by $\chi(x, y)$ for the overhang regions:

$$\chi(x,y)=\sum_{i=0}^{n}\left\{\frac{F_{si}}{2\pi\sigma_x\sigma_y}\exp\left(-\frac{1}{2}\left(\frac{(x-\mu_{xi})^2}{\sigma_x^2}+\frac{(y-\mu_{yi})^2}{\sigma_y^2}\right)\right),\right.$$
$$\left.|(x-\mu_{xi})^2+(y-\mu_{yi})^2<(\varrho_{Gi}d_{\text{self-support}})^2\right\}, \tag{5}$$

where F_{si} represents the traction force of the ith support, and ϱ_{Gi} is given by:

$$\varrho_{Gi}=\sqrt{\max(1,F_{si}/F_{\text{break}})}, \tag{6}$$

where μ_{xi} and μ_{yi} represent the coordinates of the support point.

The expansion coefficient ϱ_{Gi} adaptively modulates the support structure's strength and stability through variable force field distribution. For supports experiencing forces below the fracture threshold F_{break}, $\varrho_{Gi}=1$ setting maintains basic circular coverage with inter-support distances of approximately $2\times d_{\text{self-support}}$. Values of $\varrho_{Gi}<1$ are avoided as they lead to discontinuities in the force field between adjacent supports. When forces exceed the fracture threshold, $\varrho_{Gi}>1$ extends the stress influence range to incorporate nearby force-augmented overhang regions, as defined in Equation (6).

The goal is to precisely place the support structures according to the stress distribution of the overhang regions. To achieve this, it is assumed that the force at each location in the stress field is borne by the nearest support point, providing a simplified model for analysing the impact range of the support structures. To delineate the influence region of each support point, a Voronoi diagram is introduced. In this application, support points act as seed points of the Voronoi diagram, defining a clear influence region for each support point. For any support point p_i, its corresponding Voronoi cell region is defined by:

$$\Omega_i=\left\{p \mid \|p-p_i\|\leqslant\|p-p_j\|,\ \forall\, j\neq i,\ p\in R\right\}, \tag{7}$$

where R represents the overhang region.

The objective function aims to minimise the number of support points n, subject to two essential constraints: the mechanical stress constraint for each support point and the circular coverage constraint for all support points collectively. This can be mathematically expressed as:

$$\begin{gathered}\text{minimize}\quad n,\\ \text{subject to:}\\ \forall\, p_i,\ \sum_{p\in\Omega_i}\chi(p)\leq F_{\text{break}},\\ R_s\subseteq\bigcup_{i=1:n}\mathit{circle}(p_i,\,d_{\text{self-support}}),\end{gathered} \tag{8}$$

where R_s represents the geometric overhang region requiring support. The first constraint ensures that the accumulated stress within each support point p_i influence region Ω_i does not exceed the support's breaking threshold F_{break}. The second constraint guarantees that the union of circles centred at support points, each with radius $d_{\text{self-support}}$, completely encompasses the geometric overhang region.

Direct optimisation of this objective function poses significant computational challenges due to its combinatorial nature and multiple constraints. Therefore, we propose an iterative optimisation strategy: instead of directly minimising the number of support points, we first optimise the distribution of a fixed number of support points to satisfy both constraints as much as possible simultaneously. The optimisation process continues by incrementally adding support points until both constraints are fully satisfied. To facilitate this optimisation approach, we introduce a novel energy function to optimise the distribution of support points, defined by two objectives: first, the traction force on each support in the Z-direction should be balanced as evenly as possible; second, geometric overhang regions must meet overhang constraints. The energy function is defined as follows:

$$E_{\text{total}}=\lambda^k E_f+E_o, \tag{9}$$

where E_f represents the energy induced by the forces and E_o corresponds to the energy term for optimising the overhang constraints. To dynamically balance the importance of force and overhang constraints during the optimisation process, we introduce a decreasing coefficient λ^k:

$$\lambda^k=\lambda^{k-1}\times\varrho_k, \tag{10}$$

where ϱ_k is less than 1, ensuring that λ^k gradually decreases as iterations progress. This design divides the optimisation process into two phases:

Initial Phase: The focus is on the force constraint, helping to determine reasonable initial support positions based on force distribution.

Later Phase: The emphasis shifts to optimising overhang constraints in regions that do not initially satisfy them. For support points that already meet overhang constraints, further adjustments are made based on force constraints.

The term E_f optimises support point positions to ensure traction forces across all supports are as uniform as possible, thereby minimising the number of support points. This energy term is inspired by the energy function of centroidal Voronoi tessellation (CVT) [34]:

$$E_f = \sum_{i=1}^{n} \int_{\Omega_i} \chi(p) \left\| p - p_i \right\|^2 \mathrm{d}p, \tag{11}$$

where $\chi(p)$ denotes the density at point p. This approach allows for more effective distribution of support points, balancing the load to increase structural stability and reduce material waste.

The objective of E_o is to ensure that a circle centred at each support point with a radius of $d_{\text{self-support}}$ completely covers the geometric overhang regions. This energy term calculates the sum of the products of the furthest distance from the support point in each support point's Voronoi region and its weight ω_i:

$$E_o = \sum_{i=1}^{n} \omega_i \left(\int_{\Omega_{is}} \left\| p - p_i \right\|^{2P} \mathrm{d}p \right)^{\frac{1}{P}}, \tag{12}$$

where P approaches infinity, taking the infinity norm for $\|p - p_i\|$, and ω_i is defined as:

$$\omega_i = \begin{cases} 1, & \text{if } \left\| p - p_i \right\| > d_{\text{self-support}}, \\ 0, & \text{otherwise.} \end{cases} \tag{13}$$

When this distance is less than the maximum overhang distance $d_{\text{self-support}}$, ω_i is set to 0, indicating that the support in the current region meets the overhang constraints and thus is optimised based solely on force constraints.

The partial derivative of energy functional E_f with respect to support point position p_i was derived [34, 35] as:

$$\frac{\partial E_f}{\partial p_i} = 2m_i(p_i - b_i), \tag{14}$$

where m_i denotes the density integral over the ith Voronoi region:

$$m_i = \int_{\Omega_i} \chi(p) \, \mathrm{d}p, \tag{15}$$

and b_i represents the centroid position of the ith Voronoi region:

$$b_i = \frac{\int_{\Omega_i} p\chi(p) \, \mathrm{d}p}{\int_{\Omega_i} \chi(p) \, \mathrm{d}p}. \tag{16}$$

Based on the limit properties of L_P spaces, where the P-norm converges to the L_∞ norm as $P \to \infty$ [36], we can express the asymptotic behaviour of E_o in terms of the L_∞ norm:

$$\begin{aligned} \lim_{P \to \infty} E_o &= \sum_{i=1}^{n} \omega_i \left(\int_{\Omega_{is}} \left\| p - p_i \right\|^{2P} \mathrm{d}p \right)^{\frac{1}{P}}, \\ &= \sum_{i=1}^{n} \omega_i \max_{p \in \Omega_{is}} \|p - p_i\|^2. \end{aligned} \tag{17}$$

The partial derivative of the energy functional E_o with respect to p_i can be approximated as:

$$\lim_{P \to \infty} \frac{\partial E_o}{\partial p_i} \approx 2\omega_i(p_i - p^*), \tag{18}$$

where p^* is the point in Ω_{is} that maximises the distance from p_i:

$$p^* = \underset{p \in \Omega_{is}}{arg\,max} \, \|p - p_i\|^2. \tag{19}$$

In most optimisation processes, the farthest point p^* is unique. When multiple points in Ω_{is} share the same maximum distance from p_i, we set $p^* = p_i$ as the current solution, maintaining this position until the region Ω_{is} changes in subsequent iterations.

The partial derivative of the total energy function E_{total} with respect to position p_i can be approximated as:

$$\lim_{P \to \infty} \frac{\partial E_{\text{total}}}{\partial p_i} \approx 2\lambda^k m_i(p_i - b_i) + 2\omega_i(p_i - p^*). \tag{20}$$

Algorithm 2 Support Points Optimization Algorithm

Input: Geometric overhang region R_s, force-augmented overhang region R_w, traction force on each support F (obtained via simulation), maximum self-supporting overhang distance $d_{\text{self-support}}$, fracture threshold of the support F_{break};
Output: Support point positions S_D;
Map the support traction force to the stress field using Equation (5);
Calculate the minimum number of supports required for each region;
while True **do**
  Randomly generate initial support point locations based on the stress field;
  Optimize the distribution of support points using gradient descent, guided by Equations (9) and (20);
  if Support distribution satisfies all constraints **then**
    Record the optimized support point distribution in S_D;
    break;
  else
    Determine N_a, the maximum number of additional supports required for each overhang region due to force and overhang constraints;
    Increase the number of supports by N_a in each overhang region;
  end if
end while
Return: S_D

The specific algorithm flow is illustrated in Algorithm 2. The inputs include the geometric overhang region R_s, the force-augmented overhang region R_w, the traction force exerted on each support F, and the maximum overhang distance $d_{\text{self}-\text{support}}$, with the output being the new support point positions S_D.

First, using Equation (5), the support traction force is mapped to a stress field for the overhang regions. By calculating the adaptive ϱ_{Gi} value for each support point, the force field is moderately expanded from the geometric overhang region to the force-augmented overhang region to ensure that the support structure can effectively withstand loads.

Next, the minimum number of supports needed for each region is calculated based on two constraints: force constraints and overhang constraints:

Force constraint: The total force in each region divided by the support fracture threshold F_{break}.

Overhang constraint: The area of each zone divided by $d^2_{\text{self}-\text{support}} \times \Pi/0.9069$, where 0.9069 represents the maximum utilisation rate of circular coverage [37]. The larger value among these two determines the initial number of supports for each region. Finally, through iterative optimisation of the distribution of support points and adding extra supports as needed, the design ensures compliance with both force and overhang constraints.

Once the support contact points are all settled, we generate the support structure using the minimum spanning tree algorithm [19]. This method builds the support structure by connecting nearby support points with trusses, thus enhancing the overall stability of the structure and preventing fractures caused by mechanical vibrations during printing. At the same time, to reduce the contact region between the model and the supports, we set the cross-sectional region of the contact regions to be smaller than that of other parts of the support.

2.4. Equipment and materials

This study employs the Accufab-D1s vat photopolymerization printer, which utilises bottom-up DLP technology and offers a printable range of 144 mm × 81 mm × 160 mm. To measure the total separation force during printing, we equipped the printer with a force sensor device, as shown in Figure 12(a), following the measurement method in [31, 38]. The separation force is the sum of readings from two sensors. The measurement accuracy is ensured by rigid resin vat, which consists of a steel structure and rigid glass (under the PDMS layer). We validated the setup using a 49 N standard weight tested at five random positions on the vat, achieving less than 1% measurement deviation.

While the cohesive zone model (CZM) has been widely validated in structural engineering and interface mechanics, existing studies [31, 38–40] face technical limitations in obtaining accurate force distributions in bottom-up vat photopolymerization. Installing multiple force sensors at the bottom of the vat would obstruct UV light transmission and interfere with resin curing at sensor locations, thus disrupting the printing process. Therefore, our force sensor setup focuses on measuring the total separation force, which provides essential machine and material parameters for our simulation model.

The resin material used is DM12, with a density of $1.1 \times 10^3 \text{kg/m}^3$ and a tensile strength of 24 MPa. The resin is housed in a V5-fast cartridge, which has a surface material with a Young's modulus of 0.3 MPa and a Poisson's ratio of 0.49. The DM12 resin itself has a Young's modulus of 1000 MPa and a Poisson's ratio of 0.4.

The experiments were conducted with a pulling speed of 2 mm/s, an exposure time of 1.1 s, and a layer thickness of 0.05 mm. The computational tasks were handled by a system with a 13th Gen Intel Core i7-13700KF CPU, 32 GB of RAM, and an NVIDIA GeForce GTX 1650 GPU.

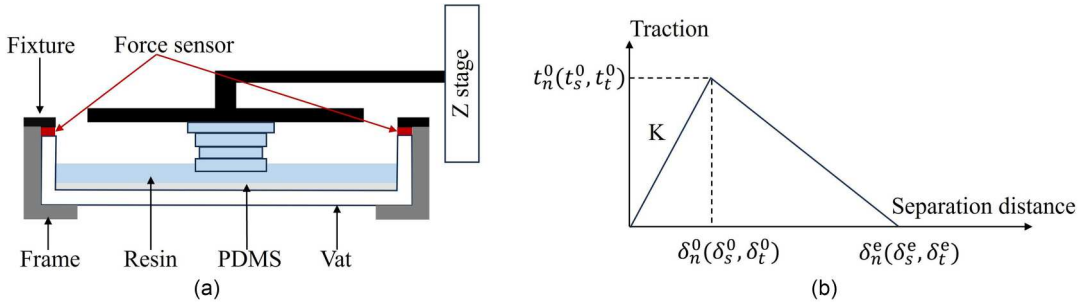


Figure 12. Schematic of the force sensor setup and traction-separation response. (a) Illustration of the force sensor apparatus. (b) Demonstration of parameter extraction for Equation (4) based on the relationship between separation force and displacement.

3. Results and discussion

3.1. Experimental parameter determination

Through a comprehensive series of experimental studies, we determined the critical parameters for simulation, which encompass the overhang distance threshold, overhang angle threshold, and support cross-sectional dimension threshold, as well as the key parameters for converting support forces into stress fields. These parameters were systematically evaluated and optimised to ensure an accurate representation of the physical printing process and structural support requirements.

It should be noted that parameter calibration through physical experiments is an unavoidable requirement for any 3D printing system, not a limitation specific to our approach. For any bottom-up photopolymerization system, basic printing parameters (overhang angle, maximum overhang distance) and separation force simulation parameters (material properties, separation force magnitude, support failure threshold) must be calibrated through physical tests. Our method only requires computing the force field and determining hyperparameters σ_x and σ_y based on these already-required parameters.

The simulation parameters were determined through experimental testing using a circular specimen with a radius of 15 mm (cross-sectional area $A_c = 706.85$ mm^2). Following the methodology illustrated in Figure 12(b), we measured a peak separation force of $F_{\max} = 31.7$ N at the damage initiation displacement $\delta_n^0 = 0.4$ mm. From these measurements, the maximum traction at damage initiation was calculated as $t_n^0 = F_{\max}/A_c = 0.045$ MPa, and the corresponding normal cohesive stiffness parameter was derived as $K = K_{nn} = t_n^0/\delta_n^0 = 0.01125$.

To determine the optimal cross-sectional dimensions for the support contact area, we conducted experiments using square cross-sections ranging from 0.1 mm to 1 mm in 0.1 mm increments. Successful printing was achieved with a side length of 0.4 mm, as shown in Figure 13. For comparison with Materialise e-Stage 7.3 and to ensure reliable support structures, a contact surface dimension of 0.5 mm × 0.5 mm was selected. In non-contact areas, the cross-section was set to 0.7 mm × 0.7 mm. The maximum load capacity of the support, defined by

$$F_{break} = \sigma_t \times A = E \times E_f \times A, \tag{21}$$

which was determined to be 1.2 N (averaged from five measurements after excluding the highest and lowest values from seven repeated tensile tests), shown in Figure 14. To establish the maximum overhang distance, we incrementally increased the distance from 0.4 mm to 4 mm, observing the print quality. A significant layering effect and edge whitening were observed when the overhang distance exceeded 2.4 mm, as depicted in Figure 15. Consequently, 2.4 mm was set as the maximum overhang distance threshold for our hardware conditions.

Although our method does not directly depend on the maximum overhang angle, we performed experiments to establish this parameter for comparison purposes. Based on prior research [41], which identified 10° as effective for photopolymerization, we determined 6° as the overhang angle threshold for our equipment, with results shown in Figure 15.

We further compared our method against Materialise e-Stage 7.3 using the same overhang distance and angle thresholds. For mechanical performance, our simulation parameters, including the maximum traction stress along the Z-direction t_n^0, were set to match the suction

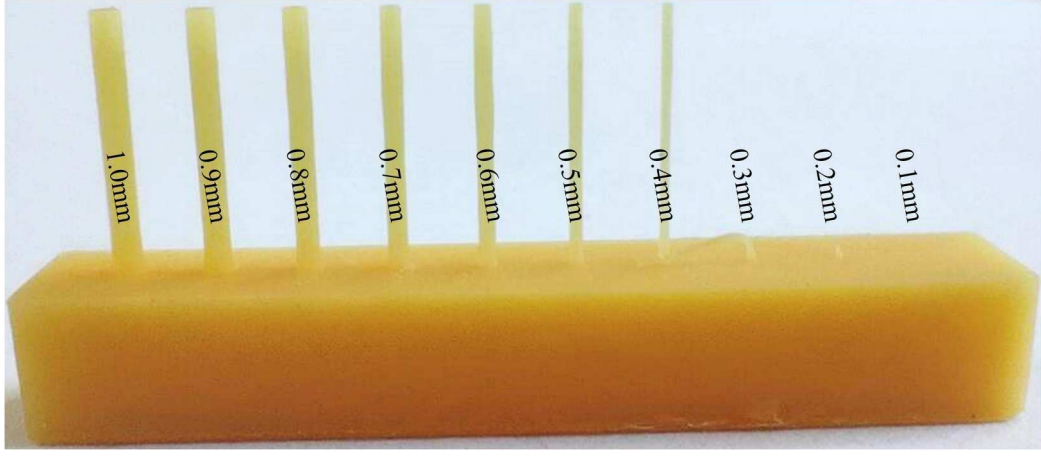


Figure 13. Experiment determining the minimum side length of the support.

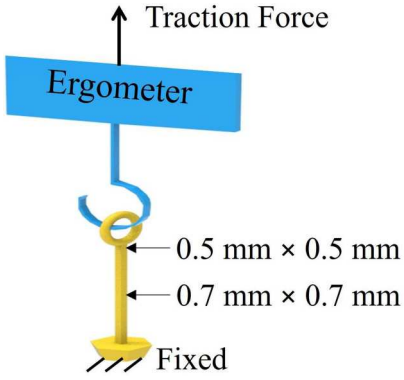


Figure 14. Schematic diagram of the support breakage threshold test. The yellow test model was printed and stretched using a blue force gauge until a fracture occurred. The breakage threshold was measured as 1.2 N for a cross-section of 0.5 mm × 0.5 mm.

pressure parameter of 45 KPa in e-Stage 7.3. Support fracture parameters, calculated via Equation (21), resulted in a maximum tensile strain of 0.48%. For simulation-based comparison purposes, to ensure controlled variable analysis and fair comparison of support forces between our method and e-Stage, we extracted support point locations from e-Stage and used identical support structures (shape) in both methods. This approach also helps avoid potential meshing challenges that might arise from e-Stage's complex support geometries in simulation. In actual printing experiments, we utilised e-Stage's native support structures for comparison printing tests. This experimental design allowed us to directly demonstrate the advantages of our optimisation method against e-Stage under real printing conditions while maintaining consistent support parameters.

Parameters σ_x and σ_y are key factors controlling the decay of stress field from support points to edges. To provide adequate support in areas under high stress, we expect the stress field intensity to be inversely correlated with the distance from support points. However, the stress fields between adjacent support points need to achieve smooth transitions at their boundaries; otherwise, support points would struggle to effectively cross the boundaries between different support regions during optimisation, likely leading to local optima.

As shown in Figure 16, through parameter testing within the interval $[0.25 \times \varrho_{Gi} d_{\text{self-support}}, 0.65 \times \varrho_{Gi} d_{\text{self-support}}]$, the stress field distributions exhibit distinct characteristics. When set to $0.25 \times \varrho_{Gi} d_{\text{self-support}}$, the stress fields are too concentrated around support points, creating distinct boundaries between adjacent supports that could trap the optimisation in local minima. At $0.45 \times \varrho_{Gi} d_{\text{self-support}}$, the stress fields achieve optimal balance – providing sufficient support strength while maintaining smooth transitions between adjacent support regions. However, at values $\geq 0.55 \times$

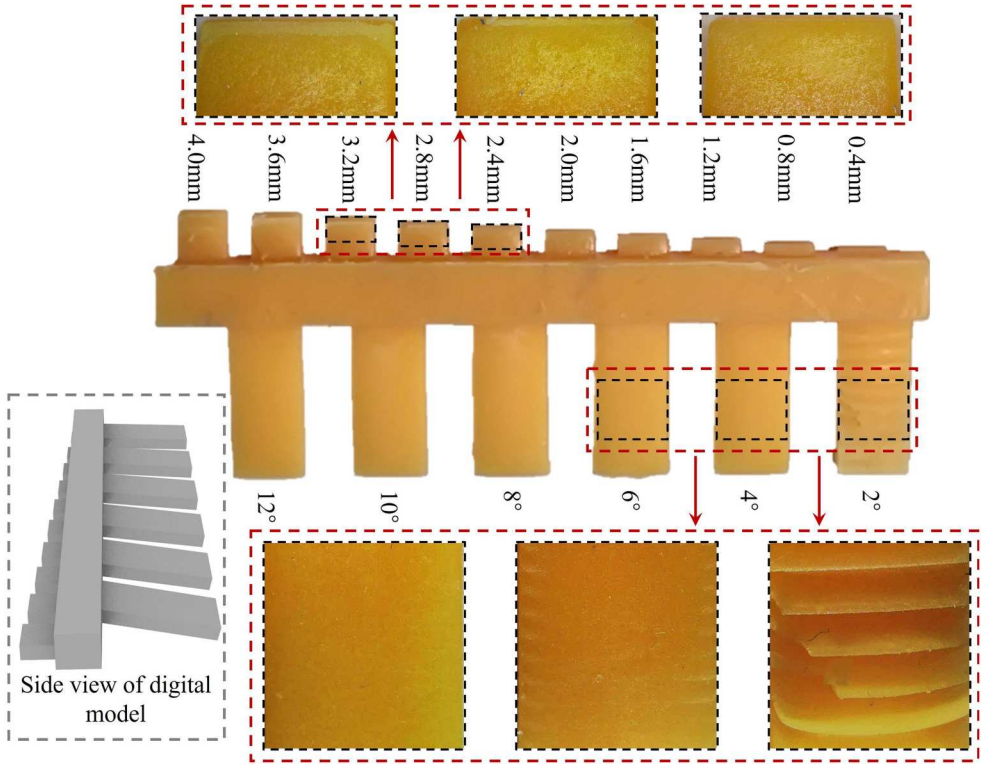


Figure 15. Experiment determining the maximum overhang distance and angle.

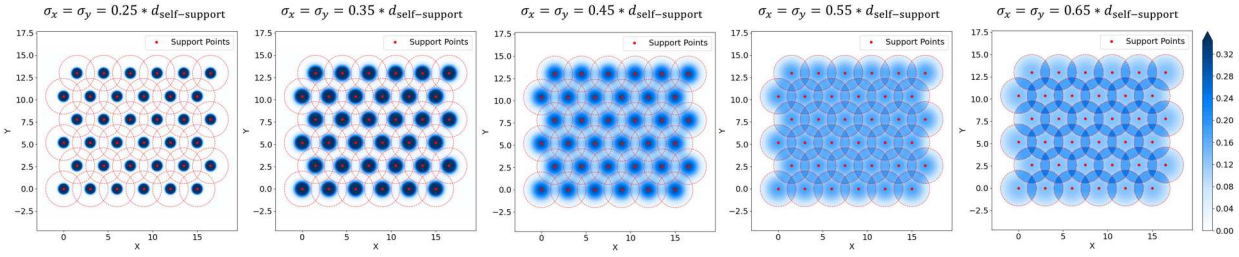


Figure 16. Stress field distributions with varying σ_x and σ_y values ($0.25 - 0.65 \times \varrho_{Gi} d_{\text{self-support}}$).

$\varrho_{Gi} d_{\text{self-support}}$, the stress fields between adjacent support points become too uniform, resulting in edge field strengths exceeding those at support points, which adversely affects optimisation convergence.

Based on this analysis, we set $\sigma_x = \sigma_y = 0.45 \times \varrho_{Gi} d_{\text{self-support}}$ as our default parameter, which ensures both adequate support strength and smooth force field transitions necessary for effective global optimisation.

3.2. Experimental results

MLP Network Outcomes. We utilised 18,000 images from various open-source datasets, such as ETH-80 [42] and Fashion-MNIST [43], to simulate separation forces along the Z-axis. These images represented a wide range of shapes, with cross-sectional areas occupying between 1% and 90% of the total pixels, as detailed in Figure 17.

The dataset was divided into training, testing, and validation subsets with a 7:1.5:1.5 ratio. The optimal architecture, consisting of three hidden layers with 64 neurones each and a learning rate of 0.001, was determined experimentally. The mean squared error (MSE) was employed as the loss function. We measured prediction accuracy by calculating the error between the predicted separation force $\overline{F}_i$ and the actual force F_i using Equation (22):

$$\psi = \frac{|\overline{F}_i - F_i|}{F_i}. \tag{22}$$

The average errors were 0.032 for the training set and 0.043 for the test set, as shown in Figure 18. The computation time for predicting the separation force was under 1 ms on our CPU.

Validation of Support Optimization. To validate our method, we tested it on models with varying structural complexities, including Dragon, hollow Cat, Armadillo, Sofa, and semi-hollow Bone models. These models ranged from solid and semi-hollow to fully hollow forms with diverse overhang areas and cross-sections.

Our analysis demonstrated the method's efficiency and reliability across these different geometries. The Dragon and Armadillo models highlighted performance on complex solids; the hollow Cat model tested adaptability to small-section hollow models; the semi-hollow Bone

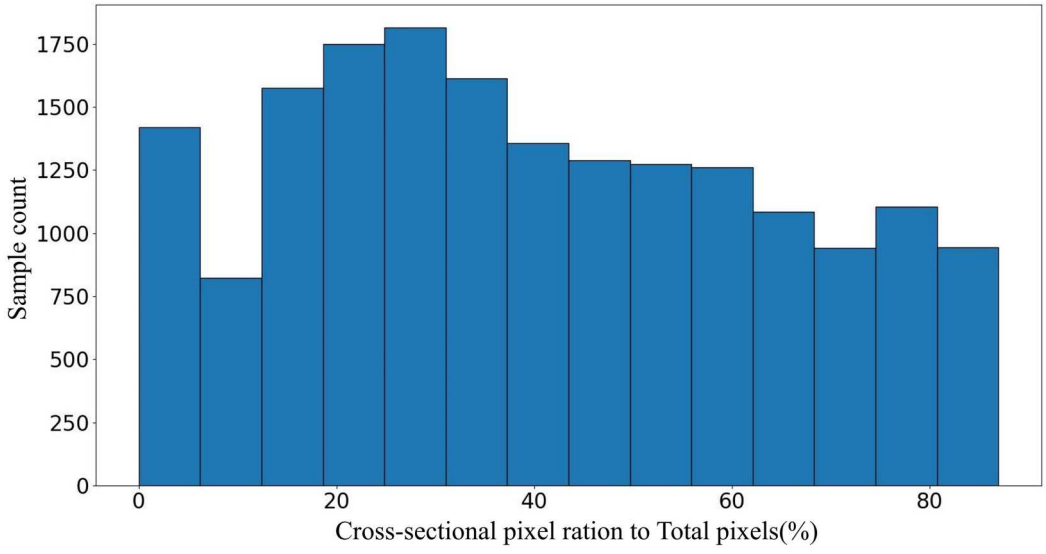


Figure 17. Distribution of cross-sectional data used in the simulation.

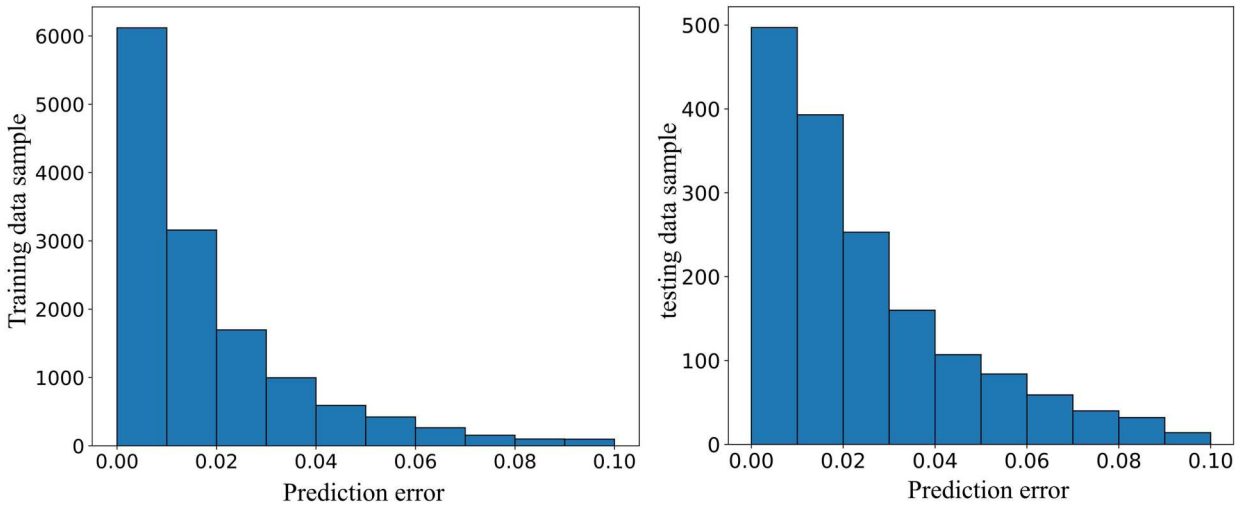


Figure 18. Prediction error distribution for training and test sets.

model demonstrated effectiveness on intermediate forms; and the Sofa model illustrated adaptability in managing small overhang areas with significant separation forces.

We present an analysis of support structure optimisation, as detailed in Table 1. Initially, the clustering was conducted on the slice images of each model, comprising hundreds to thousands of layers. By categorising these layers into approximately ten clusters, the number of required simulations was reduced by one to two orders of magnitude. The optimisation process, limited to no more than ten iterations, successfully constrained the maximum support traction force for all models within the specified threshold. Notably, the dragon model, which required the most iterations, converged after only six. The total number of simulations across all models remained below 50, confirming that our approach drastically reduces the simulation count from potentially hundreds of thousands to mere tens. Additionally, the geometric optimisation for each shape was completed in under one minute.

We conducted a comprehensive analysis of computational efficiency. The geometric optimisation phase was completed within one minute for all test models. The simulation time is primarily influenced by the model's voxel count, demonstrating a positive correlation. To illustrate this relationship, we examined two representative cases: the hollowed cat model, with its relatively low voxel count, exhibited rapid simulation convergence. In contrast, the bone model, characterised by complex geometry and higher voxel density, required substantially longer computational time. This correlation between geometric complexity and computational overhead aligns with theoretical expectations in finite element analysis.

To validate our physics simulation and geometric optimisation strategy, we observed the iterative convergence of simulation results, as depicted in Figure 19. After meeting all constraints, two additional iterations were performed to monitor changes in the maximum traction force and the number of supports. The results indicated a stable convergence trend in both metrics, with only minor fluctuations in the number of supports. In certain models, the maximum traction force slightly exceeded the threshold due to abrupt changes in the 3D geometry, leading to deviations between actual and approximated forces. However, subsequent iterations reduced this deviation, bringing the force back within the threshold, as demonstrated by the Armadillo model. The bottom panels of Figure 19 compare our method with e-Stage 7.3 results. Since e-Stage's

Table 1. Comparison of iteration metrics for different models.

	Dragon	Cat	Armadillo	Sofa	Bone
Bounding Box (mm^3)	$90 \times 63 \times 40$	$16 \times 45 \times 51$	$67 \times 80 \times 61$	$45 \times 54 \times 57$	$69 \times 90 \times 45$
Number of Sliced Layers	802	1022	1220	907	911
Number of Clusters	7	9	12	6	11
Initial Number of Supports	82	16	37	12	46
Resulting Number of Supports	170	23	89	61	159
Number of Iterations	6	3	3	2	3
Number of Simulations	42	27	36	12	33
Total simulatio time (min)	204	6	90	91	177
Geometric optimization time (min)	1	1	1	1	1

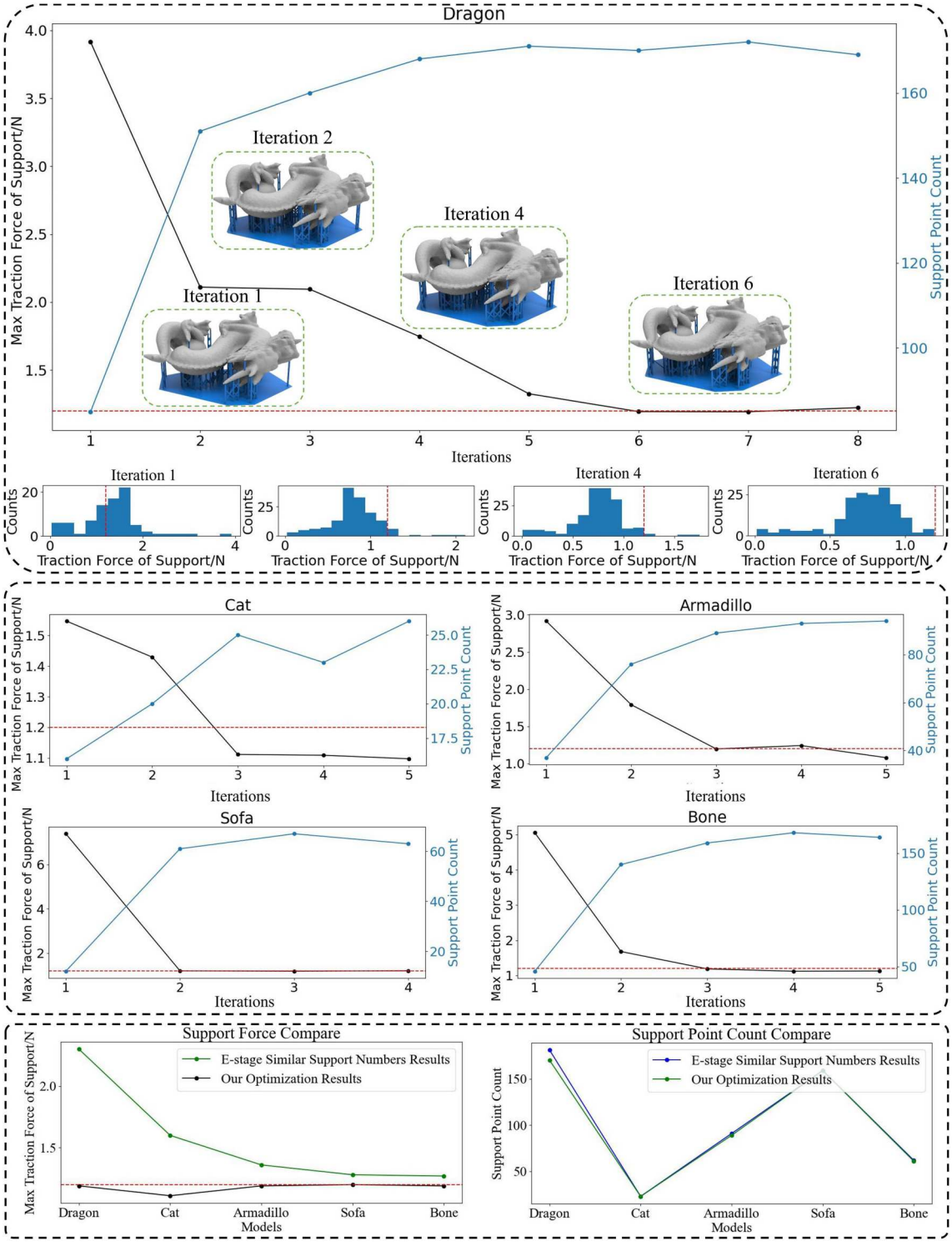


Figure 19. Iterative analysis process showing support distribution, traction forces and 3D support results for various models.

support generation is primarily controlled by overhang angle parameters, making precise support count control difficult, we conducted the comparison by setting e-Stage's support count (green line) to be equal to or slightly higher than our method's final iteration count. This comparison demonstrates that our method achieves superior force optimisation even when e-Stage uses equal or more supports, validating the effectiveness of our approach.

We further examined the distribution of traction forces on the supports, as summarised in Table 2. Our optimisation method significantly reduced the

Table 2. Traction force comparison of support optimization.

Model	Dragon	Cat	Armadillo	Sofa	Bone
Maximum Force on Initial Supports (N)	3.91	1.55	2.91	7.3	5.05
Maximum Force on Optimized Supports (N)	1.19	1.11	1.19	1.20	1.19
Average Force on Initial Supports (N)	1.36	0.78	1.8	4.46	2.42
Average Force on Optimized Supports (N)	0.74	0.54	0.85	0.96	0.84
Force Standard Deviation on Initial Supports (N)	0.63	0.50	0.63	1.17	1.46
Force Standard Deviation on Optimized Supports (N)	0.21	0.21	0.14	0.08	0.20

maximum traction forces across all models, with the most notable improvements observed in the Sofa model (from 7.3 N to 1.20 N) and the Dragon model (from 3.91 N to 1.19 N). Compared to the e-Stage, our method consistently achieves lower maximum forces (1.19–1.20 N vs. 1.89–3.42 N), demonstrating superior force control. It is important to note that the forces listed in the table represent the maximum force on individual supports. Since each model contains dozens to hundreds of supports (as shown in Figure 19), the cumulative separation force during printing can reach several hundred newtons.

The effectiveness of force redistribution is particularly evident in the standard deviation values. After optimisation, complex models like Dragon, Cat, and Bone show consistent low deviations around 0.2 N, compared to their initial high variations (0.63 N, 0.50 N, and 1.46 N respectively) and e-Stage's deviations (0.47–0.89 N). The Sofa model achieved the most uniform distribution with a deviation of merely 0.08 N, demonstrating superior force balance compared to e-Stage's 0.89 N.

The hollow cat model, despite having the lowest mean traction force (0.54 N), demonstrated our method's adaptability to different geometries. Its hollow structure required additional supports to meet overhang constraints, naturally resulting in lower average forces per support. In contrast, the sofa model achieved an average traction force of 0.96 N with a utilisation rate of up to 80%, highlighting our method's efficiency in minimising support counts while maintaining effective force distribution.

To thoroughly investigate the relationship between clustering numbers and support optimisation effectiveness, we conducted extensive experiments by varying the clustering numbers through uniform sampling of 5 points ranging from 50% to 150% of our proposed base number across five different models Figure 20.

Progressive Increase Pattern (Dragon, Cat): These models show significantly lower maximum support forces below the base clustering number (75%), followed by a sharp increase between 75-100% of the base clustering number. The force values then stabilise after reaching the base clustering number, with minimal variation up to 150%. For these complex models with varying force distributions across layers, using 50% of our proposed clustering number leads to severe underestimation of support forces, with maximum forces up to 40% lower than those obtained with our method.

Step-Jump Pattern (Armadillo, Bone): This model exhibits an immediate jump in force value at 75% of base clustering number and maintains consistent force values afterwards. This pattern reflects their moderate geometric complexity, where cross-sectional and geometric variations are less dramatic than the Progressive Increase models but still require sufficient clustering to accurately capture force distributions.

Constant Pattern (Sofa): This model exhibits consistent force values across all clustering ratios (50–150%). This equivalence occurs because the maximum support forces are concentrated in a few critical layers, where most layers are subsets of the layers with maximum cross-sections. In such cases with simple geometry, even reduced clustering numbers can accurately capture the critical supporting structures that determine the maximum forces.

These patterns demonstrate that our proposed base clustering number effectively captures the maximum support forces. Using fewer clusters (50–75%) often leads to force underestimation, particularly in complex geometries like the Dragon and Cat. Increasing clusters beyond our proposed number (100–150%) yields minimal to no

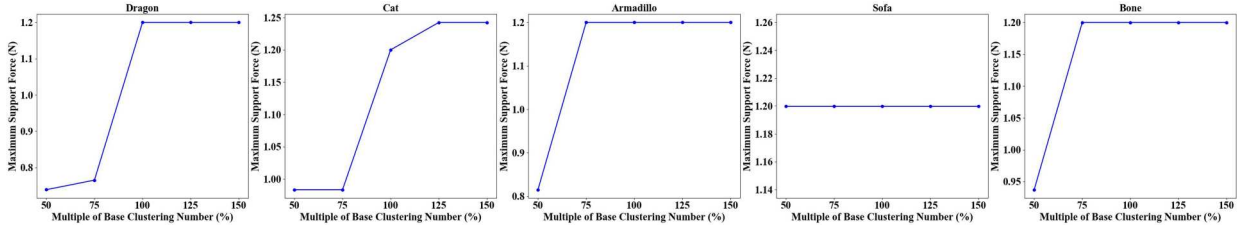


Figure 20. Maximum support forces under different clustering ratios (50%–150% of base clustering number) for five test models.

improvements in force estimation accuracy ($< 5\%$), validating that our method achieves optimal balance between computational efficiency and result accuracy. These findings conclusively demonstrate that our clustering-based approach successfully reduces simulation count while maintaining the reliability of support optimisation.

3.3. Comparison

For comparative analysis, three overhang angle configurations were investigated: 6°, 30° (default overhang threshold used in e-Stage software), and optimised angles derived through binary search coupled with CZM simulation validation using Section 2.3.1 methodology. The support structures generated by our method were benchmarked against those produced by Materialise e-Stage 7.3. Figure 21 shows that under a 6° overhang angle, the supports generated by e-Stage were insufficient, failing to expand adaptively overhang areas to meet force conditions. For instance, in the sofa model, the actual number of supports was inadequate to withstand the maximum traction force limit of 1.2 N per support, leading to potential failure under separation forces exceeding 30 N.

Further validation was conducted through physical experiments involving the printing of support structures for five different models. Accounting for inevitable simulation errors, including those related to separation forces, we adjusted the support contact section from 0.5 mm × 0.5 mm to 0.53 mm × 0.53 mm, incorporating

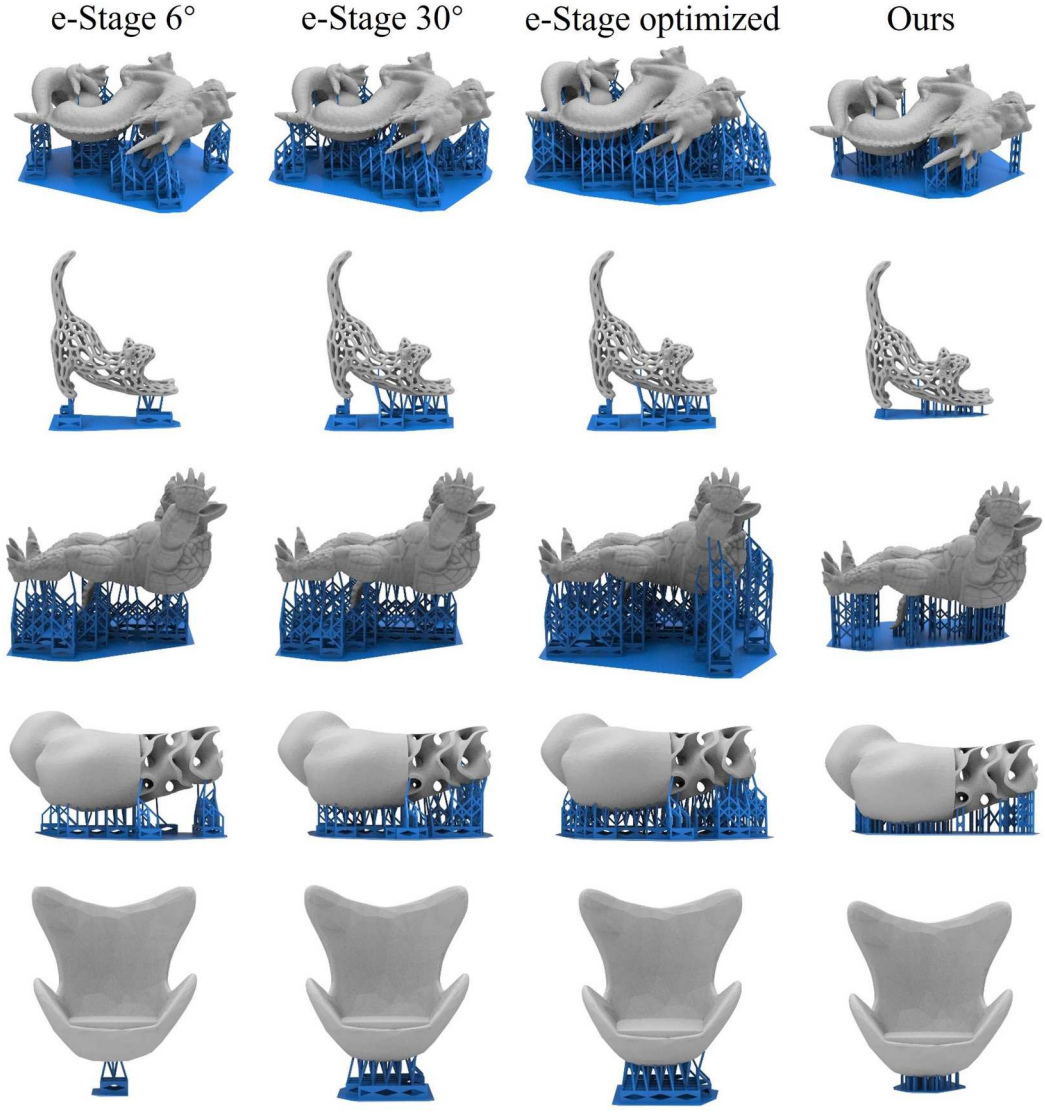


Figure 21. Comparison of support structures. From left to right, the images display support structures generated by e-Stage for overhang angles of 6°, 30°, and print-optimized angles that ensure fabrication success, respectively. The last column shows the results produced by our method.

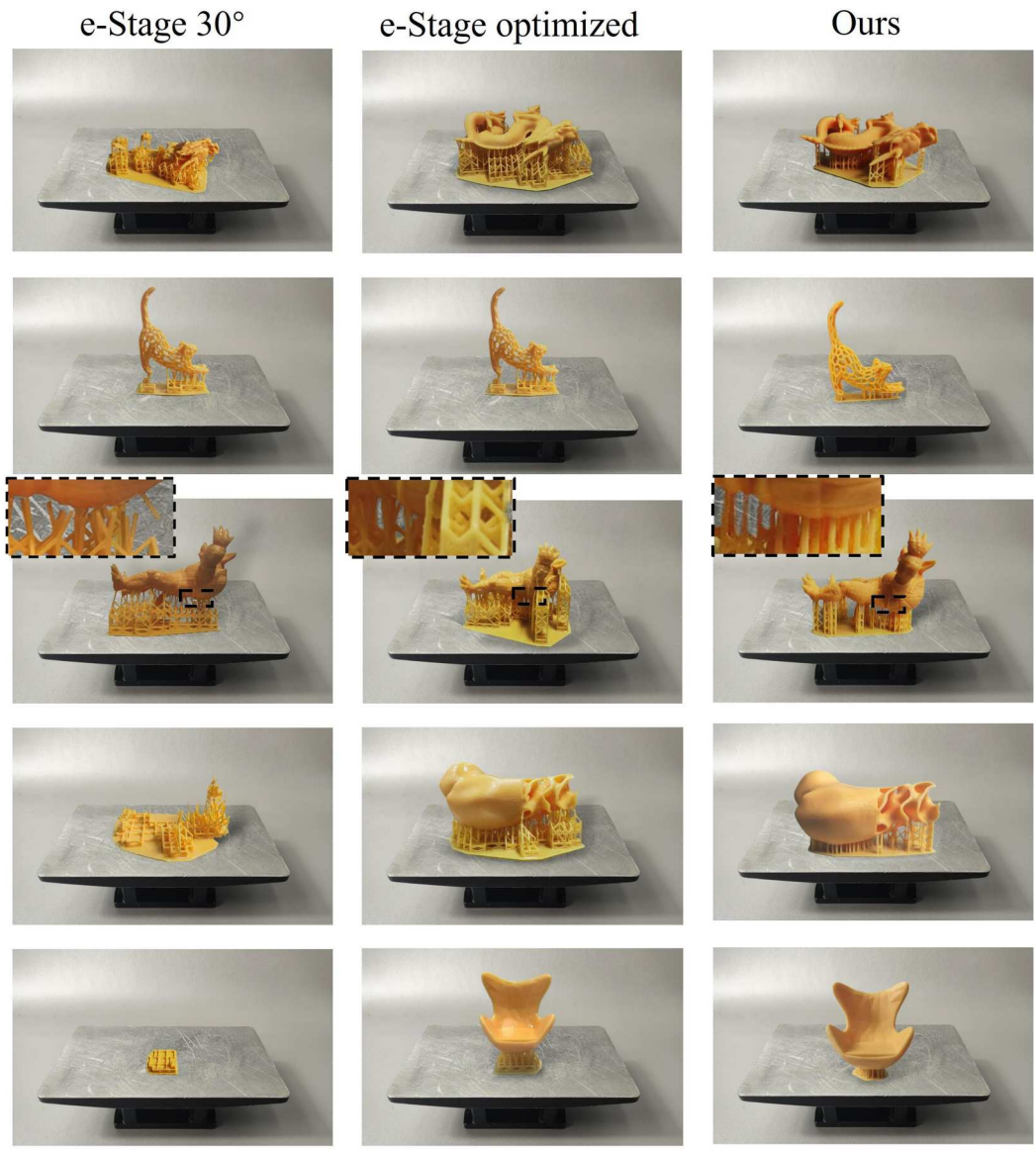


Figure 22. Comparison of printing results.

a 10% engineering margin. Additionally, each support was reinforced with a convex base to enhance adhesion to the printing platform, preventing detachment due to excessive separation forces. We conducted comprehensive fabrication experiments to evaluate the effectiveness of different support generation approaches Figure 22, Table 3. Each model was printed twice under strictly controlled environmental conditions, with consistent temperature and humidity maintained by air conditioning. All prints used identical materials and printer settings to ensure experimental consistency. The e-Stage method with a 6° overhang angle was excluded from physical testing due to evident insufficient support that would lead to printing failure. The comparison focussed on three configurations: e-Stage with 30° overhang angle, e-Stage with print-optimised overhang angles, and our proposed method.

Our experimental evaluation demonstrates the effectiveness of the proposed method through comparative studies with e-Stage. When using e-Stage with a fixed 30° overhang angle, significant printing failures were observed in the Dragon, Bone, and Sofa models, while minor defects appeared in the Armadillo model. In contrast, both our method and e-Stage with optimised angles successfully fabricated all test models. Quantitative analysis reveals that our method achieved a 32% reduction in support structures compared to the successful e-Stage prints.

The reduction in support structures directly contributes to improved surface quality. Since each support contact

Table 3. Quantitative comparison of support structure generation methods.

Evaluation Metrics	Test Models				
	Dragon	Cat	Armadillo	Bone	Sofa
Baseline I: e-Stage (30° overhang angle)					
Number of supports	123	35	65	118	42
Fabrication success[a]	✗	✓	✗	✗	✗
Number of Iterations	1	1	1	1	1
Total time consumption (min)	1	1	1	1	1
Baseline II: e-Stage (optimized parameters)					
Critical overhang angle (°)	55	30	49	41	46
Number of supports	314	35	134	181	74
Fabrication success[a]	✓	✓	✓	✓	✓
Number of Iterations	7	5	6	7	7
Total time consumption (min)	239	10	181	319	414
Our Method					
Number of supports	**170**	**23**	**89**	**159**	**61**
Fabrication success[a]	✓	✓	✓	✓	✓
Number of Iterations	6	3	3	2	3
Total time consumption (min)	205	7	91	92	178
Support reduction ratio (%)[b]	**45.9**	**34.3**	**33.6**	**12.2**	**17.6**
Time reduction ratio (%)[b]	**14.2**	**30**	**49.7**	**71.2**	**57.0**

[a]✓: fabrication successful; ✗: fabrication failure. [b]Reduction ratio calculated relative to Baseline II. *Overall support reduction: 32.0% compared to e-Stage Successful Printing.

point potentially affects the surface finish during removal, the 32% decrease in support count under identical contact radii proportionally reduces the surface area susceptible to damage (as shown in Figure 1). This indicates that our optimisation approach enhances both force distribution and post-processing surface quality through minimised support-surface interactions.

Furthermore, our method demonstrates superior computational efficiency. The e-Stage with optimised parameters (Baseline II) requires 5–7 iterations of binary search and simulation verification, consuming 10–414 minutes of computation time across different models. Our method converges in 2–6 iterations with computation times ranging from 7 to 205 minutes, achieving time reduction ratios of 14.2%–71.2%. These results collectively validate that our method outperforms e-Stage in terms of both support structure optimisation and computational efficiency.

To validate the prediction accuracy of our MLP model, we designed a controlled experiment by continuously printing the same cross-section for 30 layers. The first 20 layers were printed with larger cross-sections and longer exposure time to prevent detachment from the build platform, while the subsequent layers were used for force measurement. Similar to previous studies [44, 45], we observed inherent fluctuations (mostly within $\pm$10% of the mean value in our machine) in force sensor measurements, although the underlying mechanism remains unclear. To mitigate these measurement variations, we averaged the data over multiple layers. As shown in Figure 23, the MLP predicted separation forces closely match the experimental mean values, demonstrating the excellent prediction capability of our model.

The superior performance of our method stems from its traction-based adaptive support generation mechanism, which eliminates the need for manual overhang angle adjustments. This represents a significant advancement over traditional approaches that rely solely on fixed overhang angle thresholds. While e-Stage considers detachment forces, its inability to automatically expand overhang regions often results in redundant supports in mechanically stable areas. In contrast, our method dynamically adjusts support density and distribution based on local mechanical requirements, yielding more efficient support structures. Experimental results demonstrate substantial support reduction for models with extensive overhangs (Dragon, Cat, and Armadillo). For simpler geometries (Bone and Sofa models), where e-Stage's detachment force consideration already provides reasonable efficiency, our method achieves modest improvement (>10% support reduction). These outcomes validate the effectiveness of our mechanics-based adaptive support

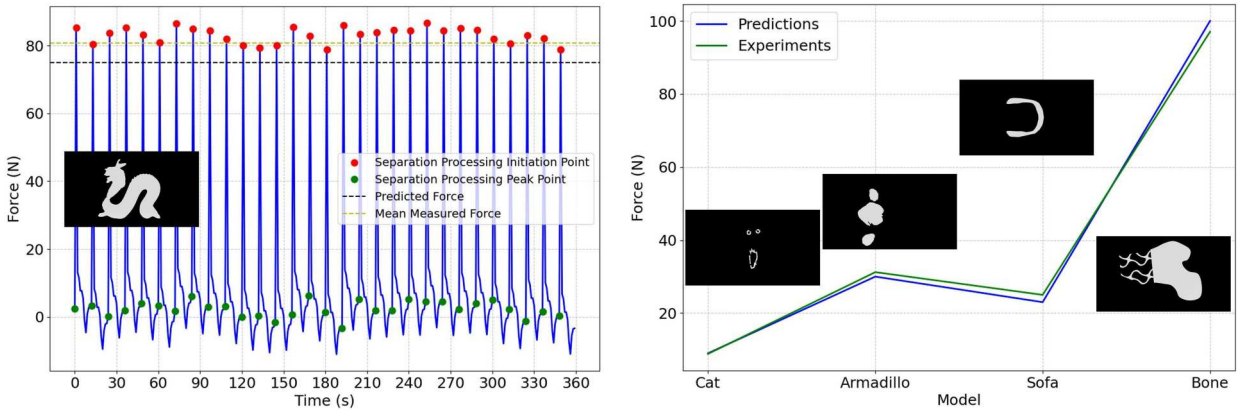


Figure 23. Comparison between MLP predictions and mean experimental forces.

generation approach, particularly for complex overhang structures.

4. Conclusion

This paper introduces an adaptive method for generating support structures in bottom-up vat photopolymerization printing. The method optimises the number and distribution of supports based on load requirements, effectively minimising the quantity of supports while satisfying load and overhang constraints.

Our layer representation extraction technique significantly reduces the required simulations by one to two orders of magnitude. By integrating physical simulations to assess separation forces, we guide and validate the optimisation of support points, reducing the number of simulations to fewer than ten. These advancements markedly lower the computational burden.

Compared to commercial software, our method offers several key advantages. It automates the generation of support structures, eliminating the need for manual adjustments by engineers and thereby enhancing production efficiency. Traditional methods and commercial software often require manual intervention to adjust overhang angles or add supports to ensure successful printing. Additionally, our intelligent support layout optimisation significantly reduces the number of supports, minimising surface damage to the model and decreasing post-processing time.

In the future, we plan to further improve simulation speed and reduce the overall time required to generate support structures. Our approach will focus on two strategies: (1) reducing the number of volumetric meshes through adaptive mesh division methods, enabling the rapid generation of high-quality mesh structures and replacing current support structures with beam elements; (2) accelerating the simulation process by employing machine learning techniques, particularly using GPUs and neural networks to solve the PDE of the separation process. The integration of these two approaches is expected to reduce simulation time to sub-second levels. With only dozens of iterations required, the entire process could be completed within one minute, demonstrating significant commercial potential.

Model orientation significantly impacts support structure requirements and printing duration. Different orientations result in varying demolding forces and corresponding support requirements. Our future research will focus on orientation optimisation to minimise support volume and printing time. Additionally, material-specific parameters, including cohesive zone model parameters and overhang angles, vary across different materials and printing systems. Therefore, extensive experimental validation is necessary for different material-printer combinations. This methodology is particularly suitable for software developers and printer manufacturers to establish material parameters for their specific systems, or to implement conservative parameters for general applications.

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Data availability

The data that support the findings of this study are available on request from the corresponding author, [Lin Lu], upon reasonable request.

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